SPINE · NEET-SS / FET / INI-SS

Lumbar disc and stenosis

1. Disc degeneration

- **Intervertebral disc: nucleus pulposus** (type II collagen, high **proteoglycan (aggrecan)** and water content; notochordal origin) and **annulus fibrosus** (type I collagen outer layers, lamellae at about 30 degrees alternating). **Largest avascular structure in the body**: nourished by diffusion through the **vertebral endplates**.
- **Degeneration:** loss of proteoglycans and water, **increased type I : type II collagen ratio**, fissures, **MMPs and cytokines (IL-1, TNF-alpha)**; smoking, genetics (strongest factor in twin studies), age.
- **MRI grading: Pfirrmann (I-V)** for disc degeneration; **Modic endplate changes:**

Modic	T1	T2	Meaning
I	Low	**High**	**Oedema/inflammation**; most associated with **low back pain**; possible low-grade infection (Cutibacterium acnes)
II	**High**	Iso/high	**Fatty marrow replacement**; commonest; stable
III	Low	Low	**Sclerosis**

2. Lumbar disc herniation

Terminology and location

- **Bulge** (over 25% of circumference) vs **herniation**: **protrusion** (base wider than the dome), **extrusion** (dome wider than base, or beyond the disc level), **sequestration** (free fragment).
- **Location: central**, **paracentral (posterolateral, commonest about 90%)**, **foraminal (far lateral, about 5-10%)**, extraforaminal.

- **Root affected:** a **paracentral L4-L5 herniation compresses the traversing L5 root**; a **far lateral (foraminal) L4-L5 herniation compresses the exiting L4 root**.
- **Commonest levels: L4-L5 and L5-S1 (about 95%)**; peak age 30-50.

Clinical

Root	Motor	Reflex	Sensory
L4	Quadriceps, tibialis anterior	**Knee jerk**	Medial leg/medial malleolus
L5	**EHL** (big toe extension), **gluteus medius** (hip abduction: Trendelenburg), tibialis posterior, ankle dorsiflexors	None reliable (medial hamstring)	**Dorsum of foot, first web space**
S1	Gastrocnemius-soleus (toe walking), peronei, gluteus maximus	**Ankle jerk**	Lateral foot, sole, little toe

- **Straight-leg raise (Lasègue)** positive between 30-70 degrees (L5/S1 roots); **crossed SLR** (pain in the affected leg when lifting the other) is **highly specific**. **Femoral stretch test** for L2-L4 (upper lumbar/foraminal herniations).
- **Red flags:** cauda equina features, progressive motor deficit, fever/weight loss/cancer history (infection/tumour), trauma, age over 50 with new pain, night pain.

Natural history and treatment

- **Favourable:** most improve within 6-12 weeks; **extruded and sequestrated fragments regress most** (resorption; inflammatory response).
- **Non-operative:** activity modification (avoid bed rest), analgesics/NSAIDs, neuropathic agents (limited evidence),

physiotherapy, **transforaminal epidural steroid injection** (short-term relief).

- **SPORT trial (Spine Patient Outcomes Research Trial):** both groups improved; **surgery gave faster and greater improvement** in pain and function (as-treated analysis), with high crossover between groups (intention-to-treat inconclusive).
- **Surgical indications: cauda equina syndrome (emergency)**, **progressive or severe motor deficit** (e.g. foot drop: earlier surgery may improve recovery), **intractable radicular pain after about 6 weeks of conservative care** with concordant imaging.
- **Surgery: microdiscectomy** (standard; interlaminar approach), endoscopic discectomy; **far lateral herniation: Wiltse paraspinal (muscle-splitting between multifidus and longissimus) approach** to avoid facetectomy.
- **Complications: recurrent herniation (about 3-7% per Campbell's; up to about 15% in some series; mostly in the first 6 months; more aggressive disc removal does not prevent it)**, **dural tear** (about 1-7%: repair primarily; headache, pseudomeningocele), wrong-level surgery, nerve root injury, **discitis**, rare **vascular injury (anterior annulus perforation: aorta/iliac vessels; hypotension: emergency laparotomy)**.

Cauda equina syndrome (CES)

- **Massive central disc (commonest cause, L4-L5/L5-S1)**, also tumour, haematoma, abscess, trauma.
- **Features: bilateral sciatica**, **saddle anaesthesia** (perianal/perineal), **bladder dysfunction (painless urinary retention with overflow is late: CES-R)**, bowel dysfunction, sexual dysfunction, reduced anal tone.
- **Classification: CES-S** (suspected: bilateral radiculopathy only), **CES-I** (incomplete: altered urinary sensation, reduced desire to void, poor

stream, but no retention), **CES-R** (retention with overflow; worse prognosis).

- **Bladder scan: post-void residual over about 200 ml** raises concern (sensitive but not specific).
- **Emergency MRI and decompression as soon as possible** (outcomes better when operated **before retention (CES-I)**; **within 48 hours** classically (Ahn meta-analysis), with current UK/GIRFT guidance for **urgent surgery at the earliest opportunity**).

3. Lumbar spinal stenosis

- **Central stenosis** (ligamentum flavum hypertrophy/buckling, facet hypertrophy, disc bulge, listhesis), **lateral recess** (subarticular; superior articular process hypertrophy compresses the traversing root), **foraminal** (exiting root; disc height loss, osteophytes). **L4-L5 commonest.**
- **Neurogenic claudication:** leg pain/heaviness/paraesthesia with walking and **standing (extension)**, relieved by **sitting or bending forward (flexion)** (**shopping-cart sign**, cycling tolerated better than walking), proximal-to-distal progression, **normal pulses**.
- **Vascular claudication:** pain in the calf with walking, **relieved by standing still** (no need to sit), absent pulses, distal-to-proximal, **cycling also painful**, low ABI.
- **Imaging:** MRI (canal cross-sectional area **under about 75-100 mm²** = stenosis; **AP diameter under 10 mm** = absolute central stenosis; **sedimentation sign**, **Schizas grading** A-D).
- **Treatment:** physiotherapy (flexion-based), analgesia, epidural injections (limited benefit in central stenosis: **LESS trial**: epidural steroid not better than lidocaine alone). **Decompression (laminectomy / laminotomy, undercutting facetectomy preserving over 50% of facets)** for persistent disabling symptoms: **SPORT stenosis cohort** and **Malmivaara** trial showed **surgery better than non-operative care**.

- **Fusion with decompression:** for **degenerative spondylolisthesis with instability**, deformity, or iatrogenic instability. **Two 2016 NEJM trials: Försth (Swedish Spinal Stenosis Study)**: **adding fusion gave no benefit** in stenosis with or without stable spondylolisthesis; **Ghogawala (SLIP)**: fusion gave **slightly better outcomes and fewer reoperations** in **stable grade I degenerative spondylolisthesis**. Result: individualise; fuse when **dynamic instability (over 3-4 mm translation or over 10-15 degrees angulation on flexion-extension)**, high disc height, facet effusion, or sagittal imbalance.
- **Interspinous spacers (X-Stop):** higher reoperation rates than decompression; limited role.

4. Spondylolysis and spondylolisthesis

Wiltse-Newman classification

Type	Name	Features
I	**Dysplastic (congenital)**	Dysplastic L5-S1 facets, **elongated pars**; high risk of progression and **cauda equina** symptoms (intact posterior arch pulls on the roots)
II	**Isthmic**	**Pars defect**: IIA **lytic (stress fracture; commonest in children/adolescents)**, IIB elongated healed pars, IIC acute fracture. **L5-S1 commonest**; **L5 root** compressed in the foramen
III	**Degenerative**	**Women over 50, L4-L5 commonest**, intact pars, facet arthropathy with **sagittal facets**; **L5 root** (lateral recess) and central stenosis
IV	Traumatic	Acute fracture of the posterior elements other than the pars
V	Pathological	Tumour, infection, bone disease

Type	Name	Features
VI	Post-surgical (iatrogenic)	After excessive decompression

- **Marchetti-Bartolozzi classification** (used in Campbell's): **developmental (high or low dysplasia, judged by L5-S1 kyphosis/slip angle and L5/S1 shape)** vs **acquired** (traumatic, post-surgical, pathological, degenerative).
- **Meyerding grade** (percentage slip of the upper vertebra on the lower): **I under 25%, II 25-50%, III 50-75%, IV 75-100%, V over 100% (spondyloptosis)**. **Low grade = I-II; high grade = III or more.**
- **Spinopelvic parameters** (see spinopelvic notes): **high pelvic incidence** predisposes to isthmic slip and progression; **slip angle** (kyphosis at L5-S1) over about 45-50 degrees predicts progression in children.
- **Spondylolysis in adolescents (athletes: gymnasts, cricket fast bowlers, footballers):**
  - Extension-related low back pain; **Scotty dog collar sign** on oblique radiographs; **SPECT/CT, or MRI (STIR oedema) for early stress reaction** (no radiation).
  - Treatment: **activity restriction (about 3 months)** ± brace; most become pain-free even if the defect does not unite. **Direct pars repair** (Buck screw, Scott wiring, pedicle screw-hook) for symptomatic L1-L4 defects/young patients without disc degeneration or slip.
- **Isthmic spondylolisthesis in children: progression is commoner in girls, before or during the growth spurt, with high-grade or dysplastic slips**. **Fusion (L4-S1 or L5-S1 in situ)** for **slip over 50%**, progression, or persistent symptoms. **Phalen-Dickson sign:** knee-flexed, hip-flexed gait with tight hamstrings.
- **Adult isthmic and degenerative spondylolisthesis:** decompression ± fusion (posterolateral, interbody: TLIF/PLIF/ALIF/LLIF). **Reduction**

of high-grade slips risks **L5 nerve root palsy** (stretch).

5. Low back pain: key evidence and syndromes

- **Non-specific low back pain:** most resolve; **avoid imaging without red flags**; stay active; NICE: exercise, manual therapy alongside exercise, CBT-based approaches; **avoid opioids/paracetamol alone**; radiofrequency denervation for facet pain in selected patients.
- **Yellow flags:** psychosocial predictors of chronicity (fear-avoidance, depression, compensation).
- **Discogenic pain/fusion:** fusion vs intensive rehabilitation trials (**Fairbank/MRC**: no significant difference; Fritzell (Swedish) showed benefit of fusion). **Disc arthroplasty** for single-level discogenic pain in select patients.
- **Facet joint pain**, **sacroiliac joint dysfunction** (diagnostic blocks), **piriformis syndrome**, **Baastrup disease** (kissing spines), **Bertolotti syndrome** (L5 transverse process pseudoarticulation with sacrum).
- **Failed back surgery syndrome:** wrong diagnosis, wrong level, recurrent herniation, epidural fibrosis, instability, infection, adjacent segment disease; **spinal cord stimulation** for neuropathic leg pain.

> Exam pearl: **Paracentral L4-L5 disc = traversing L5 root; far lateral L4-L5 = exiting L4 root (Wiltse approach). L5 = EHL and first web space; S1 = ankle jerk, lateral foot. Crossed SLR is the most specific sign.**

> Exam pearl: **Cauda equina: saddle anaesthesia, bilateral sciatica, urinary dysfunction: emergency MRI and decompression, ideally before retention (CES-I). SPORT: surgery gives faster and greater improvement for disc herniation and stenosis.**

Exam pearl: **Neurogenic claudication relieved by flexion/sitting (shopping cart, cycling ok); vascular relieved by standing still. Degenerative spondylolisthesis: women over 50, L4-L5, sagittal facets. Isthmic: L5-S1, pars defect, Scotty dog collar.** Meyerding I-V by 25% steps.

Written by Claude for Ortho preSSaleel · updated 2026-11-15. Check doubtful points in a standard textbook.